Finlay Phillips

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British Citizen — full right to work in the UK

PROFESSIONAL PROFILE

MEng Electronic and Electrical Engineering student at UCL looking for ML engineering roles in applied machine learning, model evaluation, ML infrastructure, and agent tooling.

EDUCATION

University College London 2024 â€“ 2028

MEng, Electronic and Electrical Engineering

- Coursework highlights: Data Mining and Analysis (Year 3), Digital Signal Processing and Systems (Year 3), Programming and Control Systems (75%), Mathematical Modelling and Analysis II (76%), Advanced Engineering Mathematics (Year 3).
- 3rd-Year MEng Project: Hardware-Accelerated Limit Order Book (UCL, 2026â€“2027).

Strodeâ€™s College 2022 â€“ 2024

- A-Levels: Mathematics (A*), Physics (A), Further Mathematics (A). GCSEs: eight grades 9 to 7, including English Language (9) and Mathematics (8).

TECHNICAL SKILLS

- **Languages:** Python, PyTorch, C++, Rust, TypeScript, MATLAB.
- **ML Tools and Infra:** PyTorch, ONNX Runtime, local vector stores, SQLite, PostgreSQL, Parquet, Linux/VPS, Git.
- **Web and Systems:** TypeScript, Next.js, FastAPI, REST APIs.

TECHNICAL EXPERIENCE

Co-Founder and Lead Engineer — Toolbox Maths (EdTech startup backed by University of Southampton funding) 2024 â€“ Present

- Co-founded an EdTech startup building an interactive GCSE and A-Level maths platform with practice tools, notes, and an AI study coach.
- Built automated LLM pipelines that generate structured maths questions and TikZ vector diagrams, with separate authoring, repair, and verify model stages.
- Engineered a Model Context Protocol (MCP) server layer that exposes platform tools, question banks, and interactive whiteboard state directly to AI-enabled agentic assistants.

3rd-Year MEng Project: Hardware-Accelerated Limit Order Book — UCL 2026 â€“ 2027

- Designing and implementing a low-latency SystemVerilog engine on an FPGA to parse NASDAQ ITCH protocol market data streams.
- Reconstructing and maintaining deterministic limit order book state directly in hardware at wire speed.

Tiny Latent Control Lab / VectorBot — Personal project 2025

- Created an experimental robot-control demo where plain-English commands map directly to robot movements, without generating unconstrained text as the control path.
- Took DistilGPT2 last-token activations from a fixed hook point and trained a lightweight linear probe to discrete actions such as move, reset, or toggle.
- Measured accuracy, macro F1, and abstain precision and recall, then set confidence and margin thresholds so ambiguous commands fail closed.

Identity, Local Vector Memory — Personal project 2025

- Built a local-first memory vault that securely stores personal context on-device and selectively exposes relevant facts to other applications.
- Developed a background Rust daemon that encrypts text at rest, generates local embeddings, and supports optional ONNX Runtime embedding with a preflight check.

Polymarket Market Data and Paper-Trading System — Personal project 2025

- Built a live data collection and paper-trading engine for prediction markets so strategies can be tested without risking real capital.
- Ingested live order-book depth updates into compressed Parquet on a Linux VPS with a 750 MB memory cap, serializing heavy jobs to stay inside the limit.

ACTIVITIES

- **UCL Fintech Society:** Workshops on algorithmic trading, fintech, and financial markets.